The stationary-start maintenance model and its exact delay-period predictions

Working note

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Abstract

This note proposes one change to the maintenance model — initialising the Ornstein–Uhlenbeck maintenance noise at its stationary law rather than at zero — and derives, in closed form and with complete proofs, what the two candidate models then predict for the delay-period rate of partial muscle activations. Under the current initialisation the maintenance fluctuation relaxes at rate κ, which can be as slow as $0.22\,\mathrm{s}^{-1}$, so the analysis window $[1, 3]$ s lies inside the transient and every theoretical statement is either asymptotic or conditional. Under the stationary start the relaxation rate is at least λ, uniformly in κ, and the transient is over before the window opens. Every prediction the paper needs then follows from three propositions and two theorems, with no appeal to simulation. All statements have been verified numerically; the checks are listed in Section 10.

1 The change to the model

Everything before stimulus offset is unchanged. Maintenance begins at $\tau = t_{\text{offset}} + L_{\text{motion}}$, at which time the gate-facing signal y equals the maintained target m. During maintenance,

$$\mathrm{d}\varepsilon(t) = -\kappa\,\varepsilon(t)\,\mathrm{d}t + \sigma_\varepsilon\,\mathrm{d}W_\varepsilon(t), \tag{1}$$

$$\mathrm{d}y(t) = \lambda\,[\,m + \varepsilon(t) - y(t)\,]\,\mathrm{d}t, \tag{2}$$

and $H := y - m$ is the fluctuation of the signal about the level it maintains, so that

$$\mathrm{d}H(t) = \lambda\,[\,\varepsilon(t) - H(t)\,]\,\mathrm{d}t. \tag{3}$$

Initial conditions. $H(\tau) = 0$ is *forced*: the filter output is continuous and m is defined as the value of y at τ. It is not a modelling choice and cannot be altered. By contrast $\varepsilon(\tau)$ is free. The present model sets $\varepsilon(\tau) = 0$. We propose instead

$$\varepsilon(\tau) \sim \mathrm{N}\big(0,\ \sigma_\varepsilon^2/2\kappa\big)\,, \qquad \text{independent of } W_\varepsilon, \tag{4}$$

the stationary law of (1). We call (4) the *stationary start* and $\varepsilon(\tau) = 0$ the *cold start*.

Why. ε represents fluctuations intrinsic to holding a value. Nothing makes such a process sit exactly at its mean at the instant maintenance begins, and no measurement constrains it. The cold start is therefore an unconstrained assumption, and, as Section 3 shows, it is the sole source of a slow $(1/\kappa)$ transient in a system whose only other timescale is the fast filter $1/\lambda$. Under the stationary start the model's delay-period predictions become exact rather than asymptotic; under the cold start they do not.

Remark 1. The stationary start removes the mechanism that gives Model 1 its only delay-period ramp. That is not a way of disarming Model 1: Section 6 keeps the cold start as a separate question and bounds what it can buy Model 1, showing that even its most favourable initialisation falls short of the data.

2 Exact second-order moments

Write $s^2 := \sigma_\varepsilon^2/2\kappa$, and let $\rho := t - \tau$ be elapsed maintenance time. Set

$$A_\infty := \frac{\lambda s^2}{\lambda+\kappa}, \qquad B_\infty := \frac{\lambda^2\kappa\, s^2}{\lambda+\kappa} = \frac{\lambda^2\sigma_\varepsilon^2}{2(\lambda+\kappa)}, \qquad \omega := \sqrt{B_\infty/A_\infty} = \sqrt{\lambda\kappa}. \tag{5}$$

Proposition 2 (exact moments)**.** *Under (1)–(4) with $H(\tau) = 0$, put*

$$X(\rho) := \frac{(\lambda+\kappa)\big(e^{-2\lambda\rho} - e^{-(\lambda+\kappa)\rho}\big)}{\lambda-\kappa} \qquad \big(= -2\lambda\rho\, e^{-2\lambda\rho} \ \text{ when } \lambda = \kappa\big). \tag{6}$$

Then for every $\rho > 0$, with $\sigma_H^2 := \operatorname{Var} H$, $\sigma_{\dot H}^2 := \operatorname{Var}\dot H$, $c_H := \operatorname{Cov}(H, \dot H)$,

$$\boxed{\frac{\sigma_H^2(\rho)}{A_\infty} = 1 + X(\rho) - e^{-(\lambda+\kappa)\rho}, \qquad \frac{c_H(\rho)}{A_\infty} = -\lambda X(\rho), \qquad \frac{\sigma_{\dot H}^2(\rho)}{B_\infty} = 1 + \frac{\lambda}{\kappa}\Big(X(\rho) + e^{-(\lambda+\kappa)\rho}\Big).} \tag{7}$$

In particular $\sigma_H^2(0) = 0$, $c_H(0) = 0$, $\sigma_{\dot H}^2(0) = \lambda^2 s^2$, and all three converge to (5) with $c_H(\infty) = 0$.

Proof. (ε, H) is a two-dimensional linear diffusion with drift matrix and diffusion matrix

$$F = \begin{pmatrix} -\kappa & 0 \\ \lambda & -\lambda \end{pmatrix}, \qquad Q = \begin{pmatrix} \sigma_\varepsilon^2 & 0 \\ 0 & 0 \end{pmatrix},$$

so its covariance $P(\rho)$ solves $\dot P = FP + PF^\top + Q$ with $P(0) = \operatorname{diag}(s^2, 0)$. Writing $P = \left(\begin{smallmatrix} P_{\varepsilon\varepsilon} & P_{\varepsilon H} \\ P_{\varepsilon H} & P_{HH} \end{smallmatrix}\right)$ the system is triangular:

$$\dot P_{\varepsilon\varepsilon} = -2\kappa P_{\varepsilon\varepsilon} + \sigma_\varepsilon^2, \qquad P_{\varepsilon\varepsilon}(0) = s^2, \tag{8}$$

$$\dot P_{\varepsilon H} = -(\kappa+\lambda) P_{\varepsilon H} + \lambda P_{\varepsilon\varepsilon}, \qquad P_{\varepsilon H}(0) = 0, \tag{9}$$

$$\dot P_{HH} = 2\lambda\big(P_{\varepsilon H} - P_{HH}\big), \qquad P_{HH}(0) = 0. \tag{10}$$

By (8), $\dot P_{\varepsilon\varepsilon}(0) = -2\kappa s^2 + \sigma_\varepsilon^2 = 0$, so $P_{\varepsilon\varepsilon} \equiv s^2$: *the stationary start freezes the first component exactly.* Then (9) gives

$$P_{\varepsilon H}(\rho) = \frac{\lambda s^2}{\lambda+\kappa}\Big(1 - e^{-(\lambda+\kappa)\rho}\Big) = A_\infty\Big(1 - e^{-(\lambda+\kappa)\rho}\Big),$$

and (10), integrated with $P_{HH}(0) = 0$, gives

$$P_{HH}(\rho) = 2\lambda\int_0^\rho e^{-2\lambda(\rho-v)} P_{\varepsilon H}(v)\,\mathrm{d}v = A_\infty\left[1 - e^{-2\lambda\rho} - 2\lambda\,\frac{e^{-(\lambda+\kappa)\rho} - e^{-2\lambda\rho}}{\lambda-\kappa}\right].$$

Rearranging the bracket, and using $-2\lambda = -(\lambda+\kappa) - (\lambda-\kappa)$ in its numerator,

$$P_{HH}/A_\infty - 1 = \frac{(\lambda+\kappa)e^{-2\lambda\rho} - 2\lambda e^{-(\lambda+\kappa)\rho}}{\lambda-\kappa} = X(\rho) - e^{-(\lambda+\kappa)\rho},$$

which is the first identity. Next, $\dot H = \lambda(\varepsilon - H)$, so $c_H = \lambda(P_{\varepsilon H} - P_{HH})$ and $\sigma_{\dot H}^2 = \lambda^2(P_{\varepsilon\varepsilon} - 2P_{\varepsilon H} + P_{HH})$. Substituting,

$$\frac{c_H}{A_\infty} = \lambda\Big[\big(1 - e^{-(\lambda+\kappa)\rho}\big) - \big(1 + X - e^{-(\lambda+\kappa)\rho}\big)\Big] = -\lambda X,$$

and, since $\lambda^2(s^2 - A_\infty) = B_\infty$,

$$\frac{\sigma_{\dot H}^2}{B_\infty} = \frac{\lambda^2\big[(s^2 - A_\infty) + A_\infty\big(2e^{-(\lambda+\kappa)\rho} + X - e^{-(\lambda+\kappa)\rho}\big)\big]}{B_\infty} = 1 + \frac{\lambda^2 A_\infty}{B_\infty}\big(X + e^{-(\lambda+\kappa)\rho}\big),$$

and $\lambda^2 A_\infty/B_\infty = \lambda/\kappa$. The $\lambda = \kappa$ forms follow by l'Hôpital in κ. The stated values at $\rho = 0$ and $\rho = \infty$ are read off $X(0) = 0$, $X(\infty) = 0$. □

Remark 3. σ_H rises from 0; $\sigma_{\dot H}$ *falls* from $\lambda^2 s^2 = \frac{\lambda+\kappa}{\kappa} B_\infty$, because $\dot H(\tau) = \lambda\varepsilon(\tau)$ has the noise at full amplitude while H is still at zero. So under the stationary start the position variance fills in while the velocity variance settles down.

3 How fast the transient dies

Lemma 4. *For $A, B > 0$, $A \neq B$, and $\rho > 0$,* $\left|e^{-A\rho} - e^{-B\rho}\right| \le |A - B|\, \rho\, e^{-\min(A,B)\rho}$.

Proof. Apply the mean value theorem to $x \mapsto e^{-x\rho}$ on the interval with endpoints A and B: the difference equals $|A - B|\, \rho\, e^{-\xi\rho}$ for some ξ between A and B, and $e^{-\xi\rho} \le e^{-\min(A,B)\rho}$. □

Proposition 5 (transient bounds)**.** *Set $\mu := \lambda + \min(\lambda, \kappa)$, so that $\lambda \le \mu \le \lambda + \kappa$, and*

$$\delta(\rho) := \big(1 + (\lambda+\kappa)\rho\big)e^{-\mu\rho}, \qquad \bar\gamma(\rho) := \lambda(\lambda+\kappa)\,\rho\, e^{-\mu\rho}. \tag{11}$$

Then for every $\rho > 0$ and every $\lambda, \kappa > 0$,

$$0 \le -X(\rho) \le (\lambda+\kappa)\rho\, e^{-\mu\rho}, \tag{12}$$

and consequently

$$\left|\frac{\sigma_H^2}{A_\infty} - 1\right| \le \delta(\rho), \qquad 0 \le \frac{c_H}{A_\infty} \le \bar\gamma(\rho), \qquad \left|\frac{\sigma_{\dot H}^2}{B_\infty} - 1\right| \le \frac{\lambda}{\kappa}\,\delta(\rho). \tag{13}$$

Proof. Take $A = 2\lambda$, $B = \lambda + \kappa$ in Lemma 4: $|A - B| = |\lambda - \kappa|$ and $\min(A, B) = \mu$, so $|X| \le (\lambda+\kappa)\rho\, e^{-\mu\rho}$. The sign is fixed by inspection of (6): if $\lambda > \kappa$ then $2\lambda > \lambda + \kappa$, the numerator is negative and the denominator positive; if $\lambda < \kappa$ both change sign; hence $X \le 0$ always, giving (12). The three bounds in (13) follow from (7), (12), and $e^{-(\lambda+\kappa)\rho} \le e^{-\mu\rho}$. □

Corollary 6 (relaxation is at least as fast as λ)**.** *All three moments converge to (5) at rate $\mu \ge \lambda$, uniformly in κ. In particular the $e^{-2\kappa\rho}$ mode — present in the Lyapunov spectrum $\{-2\kappa, -(\lambda+\kappa), -2\lambda\}$, and the slow mode of the cold start — has zero coefficient here, because $P_{\varepsilon\varepsilon}$ is constant.*

Remark 7 (worst case in κ). δ and $\bar\gamma$ are decreasing in ρ for $\rho \ge 1$ whenever $\lambda \ge 1$. Their worst case over κ at fixed λ is $\kappa \to 0$, which gives $\delta(1) \to (1+\lambda)e^{-\lambda}$: a bound in λ alone. The third bound in (13) carries an extra λ/κ and is therefore *not* uniform in κ; it requires a floor on κ. What it requires is weak: at $\rho = 1$ the deviation is $(\lambda/\kappa)e^{-2\lambda}$ to leading order, so a relative accuracy ϵ needs only $\kappa \gtrsim \lambda e^{-2\lambda}/\epsilon$, which at $\lambda = 10$ and $\epsilon = 10^{-2}$ is $\kappa \ge 2 \times 10^{-6}$. The floor is best read through the correlation time: at stationarity $\mathrm{Cov}(\varepsilon(t), \varepsilon(t+s)) \propto e^{-\kappa|s|}$, so $1/\kappa$ is the lag at which the maintenance noise has decorrelated to $1/e$. A process with $1/\kappa$ of minutes is constant across the 3 s delay and across dozens of consecutive trials — a session-level drift, which this model does not represent — so a floor of, say, $\kappa \ge 0.05\,\mathrm{s}^{-1}$ ($1/\kappa \le 20$ s, about seven delay periods) is a statement about what the term means, not a range imported from the simulation. Section 6 reports how the conclusions depend on it.

4 The rate of partial activations

Let s_{resp} be the responding side, g the gate, $d(t) := g - |m| - (\beta/\delta)(t - t_{\mathrm{offset}})^+$ the distance from the maintained level to the gate, and $Z := s_{\mathrm{resp}} H - d$, so that partial activations are upcrossings of zero by Z, and $\dot Z = \dot H + \beta/\delta$. The Kac–Rice formula for a differentiable Gaussian process — which requires no stationarity — gives

$$\nu(t) = p_{Z(t)}(0)\, \mathbb{E}\big[\dot Z(t) \mathbf{1}_{\{\dot Z(t) > 0\}} \,\big|\, Z(t) = 0\big] = \frac{\varphi(a(t))}{\sigma_H(t)}\, G(t), \tag{14}$$

where, writing $a := d/\sigma_H$ and $g(u) := u\Phi(u) + \varphi(u)$,

$$M = \frac{\beta}{\delta} + \frac{c_H}{\sigma_H^2}\, d, \qquad S = \sigma_{\dot{H}}^2 - \frac{c_H^2}{\sigma_H^2}, \qquad u = \frac{M}{\sqrt{S}}, \qquad G = M\Phi(u) + \sqrt{S}\,\varphi(u) = \sqrt{S}\, g(u). \tag{15}$$

Note $g' = \Phi$ and $g'' = \varphi$, so g is positive, strictly increasing and strictly convex, with $g(0) = 1/\sqrt{2\pi}$ and $g(u) \le g(0) + u$ for $u \ge 0$.

Proposition 8 (the stationary rate)**.** *Once $c_H = 0$, $\sigma_H^2 = A_\infty$ and $\sigma_{\dot{H}}^2 = B_\infty$, (14) becomes*

$$\nu = \varphi(a)\left[\frac{\beta}{\delta\sigma_H}\Phi\left(\frac{\beta}{\delta\sigma_{\dot{H}}}\right) + \omega\,\varphi\left(\frac{\beta}{\delta\sigma_{\dot{H}}}\right)\right], \qquad \omega = \sqrt{\lambda\kappa}, \tag{16}$$

whose bracket does not depend on d. Consequently:

(i) ***Model 2*** *($\beta > 0$): $a(\rho) = a(0) - \beta\rho/(\delta\sigma_H)$ falls linearly, so*

$$\frac{\mathrm{d}(\ln\nu)}{\mathrm{d}\rho} = \frac{\beta}{\delta\sigma_H}\, a(\rho), \qquad \frac{\mathrm{d}^2(\ln\nu)}{\mathrm{d}\rho^2} = -\left(\frac{\beta}{\delta\sigma_H}\right)^2, \tag{17}$$

both exact: the log rate is a downward parabola, the same one in all six conditions.

(ii) ***Model 1*** *($\beta = 0$): $M = 0$, $u = 0$, $G = \omega\sigma_H/\sqrt{2\pi}$, a is constant, and*

$$\nu = \frac{\omega}{2\pi}\, e^{-a^2/2}, \qquad \frac{\mathrm{d}(\ln\nu)}{\mathrm{d}\rho} \equiv 0. \tag{18}$$

Proof. Substitute $c_H = 0$ into (15) and (14); then $\ln\nu = \text{const} - a^2/2$ and (17) is immediate from a affine in ρ. For (ii) set $\beta = 0$: $M = 0$, $g(0) = 1/\sqrt{2\pi}$, $G = \sqrt{B_\infty}/\sqrt{2\pi} = \omega\sigma_H/\sqrt{2\pi}$, and $\varphi(a)/\sigma_H \cdot G = \omega e^{-a^2/2}/(2\pi)$, constant in ρ. □

5 Ordering across conditions

Coherence and duration enter only through m, hence only through d: by Proposition 2 the law of H depends on $(\lambda, \kappa, \sigma_\varepsilon)$ and on ρ, not on the condition, and ρ is common to all conditions because $\tau = t_{\text{offset}} + L_{\text{motion}}$ is a fixed clock time. Comparisons across conditions are therefore comparisons in d at fixed ρ.

Theorem 9 (level orders by distance, pointwise)**.** *Fix $\rho > 0$ and let $a = d/\sigma_H(\rho) \ge 1$. Then $\partial(\ln\nu)/\partial d < 0$: the rate is strictly decreasing in d, hence strictly decreasing in a. This holds for both models, transient included, and with no stationarity assumption.*

Proof. From (14), $\ln\nu = -a^2/2 - \ln\sigma_H - \ln\sqrt{2\pi} + \ln G$, and at fixed ρ only a and M depend on d, with $\partial a/\partial d = 1/\sigma_H$ and $\partial M/\partial d = c_H/\sigma_H^2$. Since $\partial G/\partial M = \Phi(u)$,

$$\frac{\partial(\ln\nu)}{\partial d} = -\frac{a}{\sigma_H} + \frac{c_H\,\Phi(u)}{\sigma_H^2\, G},$$

which is negative if and only if $d\,G > c_H\Phi(u)$. Now $G = M\Phi(u) + \sqrt{S}\varphi(u) > M\Phi(u)$ because $\sqrt{S}\varphi(u) > 0$, and

$$d\,M = \frac{\beta}{\delta}d + \frac{c_H}{\sigma_H^2}d^2 = \frac{\beta}{\delta}d + c_H a^2 \;\ge\; c_H \qquad \text{whenever } a \ge 1,$$

since $\beta, \delta, d, c_H \ge 0$. Hence $dG > dM\Phi(u) \ge c_H\Phi(u)$. □

Corollary 10. *At stationarity $c_H = 0$, and Theorem 9 holds with no condition on a: $\nu = \varphi(a) \times$ (a factor independent of d). Moreover, by (17), the log-rate slope $\beta a/(\delta\sigma_H)$ is strictly* increasing *in a. Hence, at every ρ in the window, a condition with a higher rate has a smaller a and therefore a shallower log-rate slope: level and slope order oppositely across conditions.*

6 What a non-stationary start could buy Model 1

Under Model 1, $\beta = 0$ and d is constant, so all time dependence of ν is through the three moments. Write $a_\infty := d/\sqrt{A_\infty}$ and evaluate at $\rho_0 = 1$ s (the start of the analysis window), where $\delta := \delta(\rho_0)$ and $\bar\gamma := \bar\gamma(\rho_0)$ of (11).

Theorem 11 (the transient ramp is bounded)**.** *Assume* $\lambda \geq 1$, $\delta < 1$ *and* $\eta := \dfrac{\lambda}{\kappa}\delta + \dfrac{\bar\gamma^2}{\lambda\kappa(1-\delta)} < 1$. *Then for all* $1 \leq \rho_0 \leq \rho_1$, *under Model 1,*

$$\big|\ln \nu(\rho_1) - \ln \nu(\rho_0)\big| \;\leq\; \frac{a_\infty^2\,\delta}{(1-\delta)^2} \;+\; \frac{\delta}{1-\delta} \;+\; \frac{\eta}{1-\eta} \;+\; \frac{\sqrt{2\pi}\,\bar\gamma\,a_\infty}{\omega\,(1-\delta)\sqrt{1-\eta}}. \tag{19}$$

Proof. Write $\Delta := \sigma_H^2/A_\infty - 1$, $\gamma := c_H/A_\infty$, $E := \sigma_H^2/B_\infty - 1$, so $|\Delta| \leq \delta$, $0 \leq \gamma \leq \bar\gamma$, $|E| \leq (\lambda/\kappa)\delta$ by Proposition 5, all monotonically decreasing on $\rho \geq \rho_0$ (the remark after Proposition 5). Decompose $\ln \nu = -a^2/2 - \ln \sigma_H + \ln G - \ln\sqrt{2\pi}$ and bound the change in each term over $[\rho_0, \rho_1]$. Throughout we use $|\ln(1+x)| \leq |x|/(1-|x|)$ for $|x| < 1$.

(1) The exponent. $a^2 = d^2/\sigma_H^2 = a_\infty^2/(1+\Delta)$, so

$$\tfrac{1}{2}\big|a^2(\rho_1) - a^2(\rho_0)\big| = \frac{a_\infty^2}{2}\,\frac{|\Delta_0 - \Delta_1|}{(1+\Delta_0)(1+\Delta_1)} \leq \frac{a_\infty^2}{2}\cdot\frac{2\delta}{(1-\delta)^2} = \frac{a_\infty^2\,\delta}{(1-\delta)^2}.$$

(2) The density normalisation. $\big|\ln \sigma_H(\rho_1) - \ln \sigma_H(\rho_0)\big| = \tfrac{1}{2}\big|\ln(1+\Delta_1) - \ln(1+\Delta_0)\big| \leq \tfrac{1}{2}\cdot 2\,\delta/(1-\delta) = \delta/(1-\delta)$.

(3) The speed factor. $\ln G = \tfrac{1}{2}\ln S + \ln g(u)$. For the first part, using $c_H^2/\sigma_H^2 = \gamma^2 A_\infty/(1+\Delta)$ and $A_\infty/B_\infty = 1/\omega^2 = 1/(\lambda\kappa)$,

$$\left|\frac{S}{B_\infty} - 1\right| = \left|E - \frac{\gamma^2}{\lambda\kappa(1+\Delta)}\right| \leq \frac{\lambda}{\kappa}\delta + \frac{\bar\gamma^2}{\lambda\kappa(1-\delta)} = \eta,$$

whence $\tfrac{1}{2}|\ln(S_1/S_0)| \leq \tfrac{1}{2}\cdot 2\eta/(1-\eta) = \eta/(1-\eta)$, and also $S \geq B_\infty(1-\eta) > 0$. For the second part, $M = \gamma d/(1+\Delta) \geq 0$, so $u = M/\sqrt{S} \geq 0$ and

$$u \;\leq\; \frac{\bar\gamma\,a_\infty\sqrt{A_\infty}}{(1-\delta)\sqrt{B_\infty(1-\eta)}} = \frac{\bar\gamma\,a_\infty}{\omega(1-\delta)\sqrt{1-\eta}} =: u_{\max}.$$

Since g is increasing with $g(0) = 1/\sqrt{2\pi}$ and $g' = \Phi \leq 1$, we have $g(0) \leq g(u) \leq g(0) + u$ for $u \geq 0$, so both $g(u_1)/g(u_0)$ and its reciprocal are at most $1 + u_{\max}/g(0) = 1 + \sqrt{2\pi}\,u_{\max}$, giving $|\ln(g(u_1)/g(u_0))| \leq \ln(1+\sqrt{2\pi}u_{\max}) \leq \sqrt{2\pi}\,u_{\max}$.

Summing (1)–(3) gives (19). □

Remark 12 (what the two conditions mean). Neither $\delta < 1$ nor $\eta < 1$ is an assumption about the world; both are the conditions under which the bound is finite, and each has a direct reading. δ is the relative error on σ_H^2 at the start of the window, so $\delta < 1$ says only that the position variance is known to better than 100% there — at $\lambda = 10, \rho_0 = 1$ it is 5×10^{-4}, and it never binds. η is the relative error on S, the conditional variance of the crossing speed, made of the σ_H^2 error $(\lambda/\kappa)\delta$ and the c_H^2 correction; $\eta < 1$ says S is bounded away from zero, so that $\sqrt{S}$ and $\ln S$ are controlled. Since $\eta \approx (\lambda/\kappa)\delta$, the condition $\eta < 1$ *is* the κ floor of Proposition 5, written explicitly: at $\lambda = 10$ it reads $\kappa \gtrsim 5 \times 10^{-3}$.

Corollary 13 (numerical form)**.** *Take* $\rho_0 = 1$ *s,* $a_\infty \leq 3.3$ *(Section 7), and* $\lambda \geq 10$. *Then Theorem 11 gives the following ceiling* B *on Model 1's total log-rate change across the* 2 *s window, as a function of the floor imposed on* κ*:*

$\kappa \geq$	$1/\kappa \leq$	η	B	*implied* $\lvert slope \rvert$ (s^{-1})
0.008	125 *s*	0.62	1.85	0.93
0.01	100 *s*	0.50	1.15	0.58
0.05	20 *s*	0.096	1.65×10^{-1}	8.2×10^{-2}
0.22	4.5 *s*	0.019	4.48×10^{-2}	2.2×10^{-2}
1	1 *s*	0.002	9.20×10^{-3}	4.6×10^{-3}

The observed log-rate slope is $+0.867\,\mathrm{s}^{-1}$. *At* $\kappa \geq 0.05$ — *a maintenance fluctuation that decorrelates within about seven delay periods — the ceiling is* 11 *times below the observation; at* $\kappa \geq 0.22$, 39 *times below. Increasing* λ *has the same effect: at* $\kappa \geq 0.01$ *the ceiling falls to* 6.4×10^{-2} *at* $\lambda = 12$ *and* 4.4×10^{-3} *at* $\lambda = 15$. *The* κ *floor and the* λ *prior are therefore substitutes, and only one of the two needs to be argued.*

7 The standardised distance is measured, not assumed

Theorem 11 needs an *upper* bound on a_∞. It is supplied by the data, not by a prior: inverting (18), Model 1's own stationary rate formula, gives

$$a = \sqrt{2 \ln\Big(\frac{\omega}{2\pi\nu}\Big)}, \qquad \omega = \sqrt{\lambda\kappa}. \tag{20}$$

Using Model 1's formula to pin a while bounding Model 1 is legitimate. With the observed delay-period rates 0.030–0.135 s^{-1} and (λ, κ) anywhere in the sampled range, (20) gives $a \leq 3.3$. This is the rate–ramp trade-off in its sharpest form: the rate *measures* the standardised distance, and the measured distance is too small to support the observed slope.

8 Predictions to be tested

The delay-period observations the paper tests are, in the notation of the working recap: **(L1)** the level rises with stimulus duration; **(L2)** the level rises with motion coherence; **(L3)** the coherence effect on the level is larger after the long stimulus; **(S1)** the log-rate slope is positive, $+0.867\,\mathrm{s}^{-1}$; **(S2)** it is shallower after the long stimulus, $0.896 \to 0.723$; **(S3)** it is shallower at high coherence, $0.871 \to 0.726$; **(C1)** the fitted quadratic coefficient is $c_2 = -0.0405\,\mathrm{s}^{-2}$, not distinguishable from zero. Write B for the ceiling of Corollary 13 and $T = 2$ s for the window length. A profile whose total range is at most B has, on the 20 bin centres of the window, $\lvert\text{slope}\rvert \leq 0.752\,B$ and $\lvert c_2 \rvert \leq 1.458\,B$.

Corollary 14 (L1, L2: the level)**.** *Both models predict that duration and coherence raise the level. Duration and coherence both raise the fraction of trials whose maintained target has reached the cap* θ, *hence lower* d *in the sense of first-order stochastic dominance; by Theorem 9 the rate is strictly decreasing in* d *at every* ρ, *and averaging over trials preserves the ordering.* Not discriminating.

Corollary 15 (S1: the slope)**.** *Model 1 predicts a log-rate slope of exactly zero (Proposition 8(ii)), and Theorem 11 bounds what the residual transient can add:* $\lvert slope \rvert \leq 0.752\,B$, *which is* $1.24 \times 10^{-1}\,\mathrm{s}^{-1}$ *at* $\kappa \geq 0.05$ *and* 3.4×10^{-2} *at* $\kappa \geq 0.22$. *Model 2 predicts* $\beta a(\rho)/(\delta\sigma_H) > 0$, *its only source of a ramp.* Discriminating: the observed $+0.867\,\mathrm{s}^{-1}$ exceeds Model 1's ceiling by a factor of 7 to 26.

Corollary 16 (S2, S3: the slope across conditions)**.** *Model 1 predicts the slope to be zero in* every *condition, so it predicts a condition difference of zero, with* $\lvert difference \rvert \leq 1.5\,B$. *Model 2 predicts a difference of definite sign: by Proposition 8(i) the slope is* $\beta a/(\delta\sigma_H)$, *increasing in* a,

while by Theorem 9 the level is decreasing in a; hence any manipulation that raises the level lowers the slope. By Corollary 14 duration and coherence raise the level, so both must lower the slope. Discriminating: the observed differences, 0.173 and 0.145, exceed Model 1's ceiling (0.25 at $\kappa \geq 0.05$, 0.067 at $\kappa \geq 0.22$) at the tighter floor, and their sign is the one Model 2 requires.

Corollary 17 (C1: the curvature)**.** *Model 1 predicts $c_2 = 0$ exactly, with $|c_2| \leq 1.458\,B$. Model 2 predicts*

$$c_2 = -\tfrac{1}{2}\left(\frac{\beta}{\delta\sigma_H}\right)^2 = -\frac{1}{2}\left(\frac{slope}{a}\right)^2, \tag{21}$$

constant in ρ and identical in all six conditions. With the observed slope and $a \leq 3.3$ this is $c_2 \geq -0.034$; the observed $c_2 = -0.0405 \pm 0.0416$ is consistent with both. Not discriminating at the present precision. *Under the cold start the curvature did discriminate, through the decay Model 1's variance filling forces; under the stationary start that role passes to S2 and S3, and the curvature becomes a quantitative check on Model 2 and the identification below.*

Corollary 18 (identification)**.** *Under Model 2 the three delay-period observables determine three model quantities, with no free parameter. From (17),*

$$\frac{\beta}{\delta\sigma_H} = \sqrt{-2c_2}, \qquad a = \frac{slope}{\sqrt{-2c_2}}, \tag{22}$$

and then the level fixes $\omega = \sqrt{\lambda\kappa}$ through (16). With slope $= 0.867$, $c_2 = -0.0405$ and $\nu = 0.070\,\mathrm{s}^{-1}$ this gives $\beta/(\delta\sigma_H) = 0.285\,\mathrm{s}^{-1}$, $a = 3.05$ and $\omega = 45\,\mathrm{s}^{-1}$. The value of a agrees with the independent estimate from the level alone (Section 7, $a \leq 3.3$), which is a genuine consistency check since the two routes use different observables. The value of ω is a prediction: it requires $\lambda\kappa \approx 2.0 \times 10^3$, above the range the present sweep samples ($\omega \leq 38$). Caveat: c_2 is not significantly different from zero, so (22) is a point estimate whose upper interval on a, and hence on ω, is unbounded.

9 Summary of the predictions

Observation	Model 1	Model 2
level rises with duration	yes (Thm 9)	yes (Thm 9)
level rises with coherence	yes (Thm 9)	yes (Thm 9)
coherence × duration on level	negative	negative
log-rate slope > 0	$\equiv 0$ (Prop. 8(ii))	$\beta a/(\delta\sigma_H) > 0$
slope shallower where level higher	—	yes (Cor. 10)
log-rate curvature	$\equiv 0$	$-(\beta/\delta\sigma_H)^2$

Section 8 states each row as a corollary with its numbers. The third row is not proved here: it rests on the saturation argument of the main text — the maintained target is capped pathwise at θ, so the coherence levels converge as the fraction of capped trials tends to one, and a longer stimulus raises that fraction — which concerns the distribution of m, not the moments of H, and is therefore unaffected by (4). It is combined with Theorem 9, which converts an ordering of d into an ordering of ν at every ρ.

Two rows discriminate, and both are exact statements with no residual term: Model 1 predicts a delay-period rate that is *flat*, in level and in shape, at every parameter setting.

10 Numerical verification

Every statement above was checked before being written.

Claim	Test	Result
Prop. 2	closed forms vs. numerical integration of $\dot{P} = FP + PF^\top + Q$	max rel. err. 8.9×10^{-12}
Prop. 2	the three identities in terms of X	agree to 1.5×10^{-13}
Prop. 5	$0 \le -X \le (\lambda+\kappa)\rho e^{-\mu\rho}$, exact arithmetic	worst ratio 1.000000
Prop. 5	$\lvert\sigma_H^2/A_\infty - 1\rvert \le \delta$	worst ratio 1.000000
Prop. 5	$c_H/A_\infty \le \bar{\gamma}$	worst ratio 1.000000
Prop. 5	$\lvert\sigma_{\dot{H}}^2/B_\infty - 1\rvert \le (\lambda/\kappa)\delta$	worst ratio 0.997226
Thm 11	bound vs. exact $\lvert\Delta \ln \nu\rvert$, 7435 points	0 violations, worst ratio 0.338

Grids: $\lambda \in [0.3, 60]$, $\kappa \in [0.01, 60]$, $\rho \in [0.05, 6]$, $a_\infty \in \{1, 2, 3, 4.34, 8\}$; the X bounds were verified in 50-digit arithmetic because the direct combination $P_{\varepsilon\varepsilon} - 2P_{\varepsilon H} + P_{HH}$ cancels catastrophically in double precision.

11 Consequences for the paper

Adopting (4) would remove, from the appendix, the convolution kernel K and the integrals $\int K^2$, $\int K'^2$, the auxiliary $W = K' - K^3/2\sigma_H^2$, the log-concavity lemma, the non-stationary derivative of $\ln \nu$, and the transient content of the present Corollaries A.2 and A.3. What remains is Propositions 2–5, the Kac–Rice formula (14), Proposition 8, and Theorems 9–11. The cold start becomes a single question — *could a non-stationary start rescue Model 1?* — answered by Corollary 13.

The cost is that theory and simulation must agree, so the 1200-draw parameter sweep, the admissibility screen, and every effect size would have to be recomputed under (4).